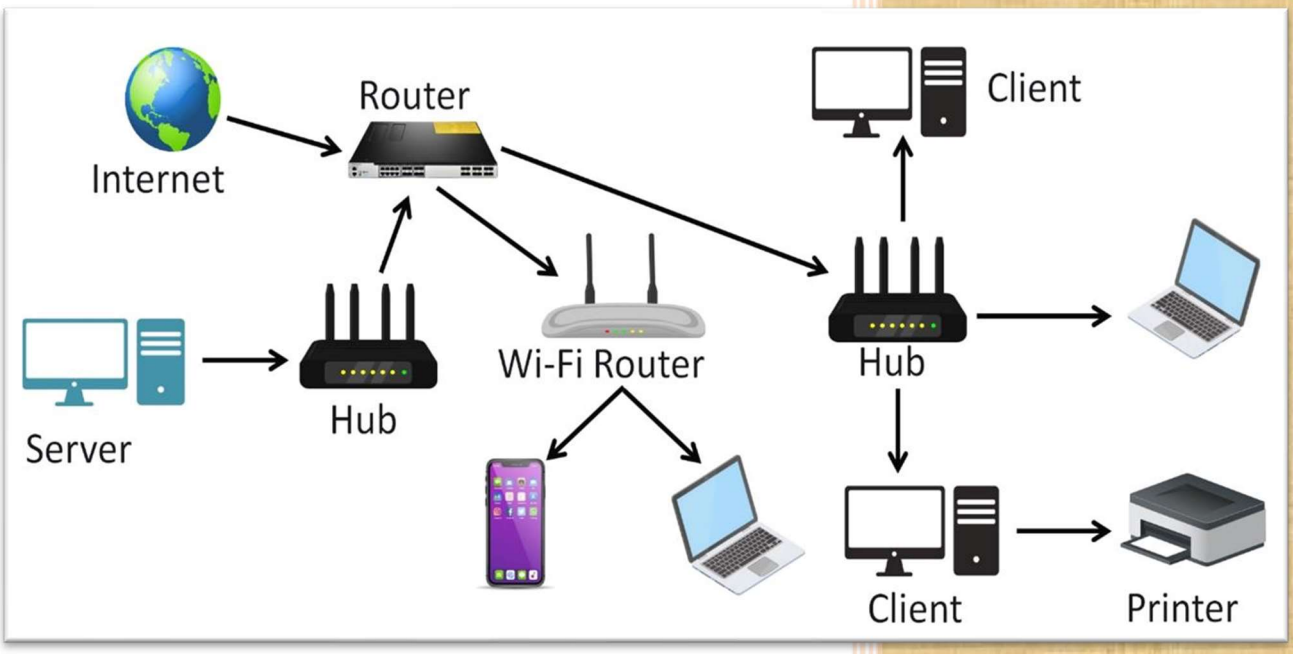
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RIZVI EDUCATION SOCIETY'S

Rizvi College of Engineering

Department of Computer Engineering

Computer Network Lab Manual



NAME:

CLASS:

RIZVI EDUCATION SOCIETY'S

Rizvi College of Engineering

Department of Computer Engineering

Lab File

Class & Semester: *TE (Sem 5)*

Subject: *Computer Network Lab*

Name: ……

Roll No.: ……..

UIN: ……

Document Revised on: 09*/ 07 / 2026*

RIZVI EDUCATION SOCIETY'S

Rizvi College of Engineering

Department of Computer Engineering

Certificate

This is to certify that…… of Third Year Computer Engineering Department has performed all the experiments of Computer Network with satisfactory results.

……..

Subject In-charge

INSTITUTE VISION

To become a leading entity in transforming the diverse class of learners into innovators, analyzers, and entrepreneurs competent to develop eco-friendly sustainable solutions and work in multi-disciplinary environment to meet the global challenges and contribute towards nation building.

INSTITUTE MISSION

IM1. **Impart Core Fundamental Principles**: To impart core fundamental principles of engineering and science, through conventional and innovative methods which will help the learners acquire skillset to create solutions to complex engineering problems.

IM2. **Bridge the Technical Skill Gap**: To bridge the industry – academia gap through curriculum enrichment activities for industry readiness.

IM3. **Inculcate Professional Etiquettes and Ethics**: To groom the learners through well-planned training & placement activities laced with professional etiquettes and ethics leading to their holistic development thus enabling them to acquire distinguished positions in prominent organizations and inculcating lifelong learning capability.

IM4. **Research and Development**: To provide modern infrastructure and the necessary resources, for planning and implementing innovative research ideas, leading to entrepreneurship.

DEPARTMENT VISION

To become a center of excellence in the field of Computer Engineering to transform diverse class of learners into skilled professionals with ethical values capable of contributing towards nation building.

DEPARTMENT MISSION

DM1. **Quality Technical Education**: To provide quality technical education with the help of modern resources for finding solutions to complex problems of computer engineering.

DM2. **Leadership & Entrepreneurial skills**: To inculcate moral and ethical values while acquiring leadership, entrepreneurial skills and overall personality development.

DM3. **Professional Skills and Lifelong Learning**: To inculcate professional skills and lifelong learning for further education as well as acquiring distinguished positions in leading software industries.

DM4. **Research and Development**: To encourage creative thinking for planning and implementing innovative ideas leading to meaningful research and development considering economical, societal and environmental factors.

PROGRAM OUTCOMES

PO1. Engineering knowledge
PO2. Problem analysis
PO3. Design/development of solutions
PO4. Conduct investigations of complex problems
PO5. Modern tool usage
PO6. The engineer and society
PO7. Environment and sustainability
PO8. Ethics
PO9. Individual and team work
PO10. Communication
PO11. Project Management and finance
PO12. Life-Long Learnings

PROGRAM SPECIFIC OUTCOMES

PSO1 Open-Source Tools: To encourage the students to work using open-source software's & tools in diversified areas of computer science.

PSO2 Industry Readiness: To enable the students to acquire the necessary skill set required to develop, test, install, deploy, and maintain a complete software system for business and other applications, that makes them industry ready.

PROGRAM EDUCATIONAL OBJECTIEVES

PEO1. **Successful Career**: To build a successful career in leading industries related to the field of Computer Engineering wherein the engineer will be able to provide the necessary solutions to the challenges witnessed in existing and new business models.

PEO2. **Leadership Qualities**: To exhibit the qualities of team spirit, leadership, and problem-solving skills to achieve top positions in the organization or to enhance entrepreneurial skills.

PEO3. **Adaptability to New Technology**: To be able to adapt to new technologies and platforms and share their knowledge with their peers in the allied fields.

PEO4. **Research & Higher Studies**: To develop proficiency in computer engineering and related fields to be able to work in multi-disciplinary areas with a strong focus on innovation and research.

Lab Objectives

To provide:

1) To learn various simulator to implement and analyze the network in it
2) To learn various TCP/IP utility command.
3) To learn different network topologies using appropriate routing algorithms
4) To learn different protocols in transport layer stack
5) To implement basic terminal security commands on GNS3 to secure network devices and analyze captured network packet.
6) To implement basic network security configurations and analyze captured packets to identify potential threats

Lab Outcomes

Learner will be able to…

CO1 :	**Compare Various network simulator.**
CO2 :	**Analyze TCP and UDP data traffic for network performance evaluation.**
CO3 :	**Design small network topologies using appropriate routing algorithms.**
CO4 :	**Demonstration of transport layer protocols on Wireshark/NS2**
CO5 :	**Develop a program for Client-Server architecture using socket programming**
CO6 :	**Identify security threats by analyzing captured network packets**

Rubrics

The following rubrics will be used to assess the work submitted by the students.

1. **For Experiment 1 to 10**

Criteria 1: Delivery:			
Has the student Submitted experiments in stipulated time			
3	**2**	**1**	**0**
Excellent	Good	Average	Poor
Submitted before the due date.	Submitted after 1 week of due date	Submitted after 2 weeks of due date	Submitted after 3 weeks of due date
Criteria 2 : Pre Lab & Post Lab Question Answered	How well the topic is understood to the students based on question asked to them.		
3	**2**	**1**	**0**
Excellent	Good	Average	Poor
Answered all questions with appropriate answers	Answered few questions with appropriate. answers.	Answered only 1 question with appropriate. answers.	Answered zero Questions
Criteria 3: Simulation Output Screenshot Attached With Name & UIN			
3	**2**	**1**	**0**
Excellent	Good	Average	Poor
Proper Required output Screenshot with Name & UIN is attached in the experiment with all the parameters.	Required output Screenshot is attached in the experiment with some parameters missing	Improper output Screenshot with some missing parameters	Improper output with all the missing parameters required for the Output.

2. For Assignments 1 to 2

Criteria 1 : Delivery			
Submitted assignments in stipulated time			
3	**2**	**1**	**0**
Excellent	Good	Average	Poor
Submitted before the due date.	Submitted after 1 week of due date	Submitted after 2 weeks of due date	Submitted after 3 weeks of due date
Criteria 2: Understanding			
How well the Concept is understood:			
3	**2**	**1**	**0**
Excellent	Good	Average	Poor
Answered 3 questions with appropriate answers	Answered 2 questions with appropriate. answers.	Answered only 1 question with appropriate. answers.	Answered zero were not satisfactory.
Criteria 3: Diagrams			
Has the student drawn the diagrams with proper labelling & neatness & visibility of words:			
3	**2**	**1**	**0**
Excellent	Good	Average	Poor
Proper representation of diagram as per reference / Text book are produced for each question with proper labeling & neatness & proper visibility of words.	Proper representation of diagram as per reference / Text book are produced for each question but with improper visibility of words	Improper representation of diagram as per reference / Text book are produced for each question	Diagram is missing from the questions

Lab Guidelines

● Guidelines for performing the Experiments:

1. Students are advised to come to the laboratory at least 5 minutes before (to starting time), those who come after 5 minutes will not be allowed into the lab.

2. Plan your task properly much before to the commencement, come prepared to the lab with the synopsis/ program / experiment details.

3. Student should enter into the laboratory with: a. Laboratory observation notes with all the details (Problem statement, Aim, Algorithm, Procedure, Program, Expected Output, etc,) filled in for the lab. session. b. Laboratory Record updated up to the last session experiments and other utensils (if any) needed in the lab. c. Proper Dress code and Identity card.

4. Sign in the laboratory login register, write the TIME-IN, and occupy the computer system allotted to you by the faculty.

5. Execute your task in the laboratory, and record the results / output in the lab observation notebook, and get certified by the concerned faculty.

6. All the students should be polite and cooperative with the laboratory staff, must maintain the discipline and decency in the laboratory.

7. Computer labs are established with sophisticated and high-end branded systems, which should be utilized properly.

8. Students / Faculty must keep their mobile phones in SWITCHED OFF mode during the lab sessions. Misuse of the equipment, misbehavior with the staff and systems etc., will attract severe punishment.

9. Students must take the permission of the faculty in case of any urgency to go out ; if anybody found loitering outside the lab / class without permission during working hours will be treated seriously and punished appropriately.

10. Students should LOG OFF/ SHUT DOWN the computer system before he/she leaves the lab after completing the task (experiment) in all aspects. He/she must ensure the system / seat is kept properly

INDEX

Sr. No.	Title	Page No.	Performed On	Sign & Date	Rubrics Points
1	Exp 1:To Implement and analyze TCP/IP utility commands.				
2	Exp 2: To Configure network devices, such as hubs and switches within a network topology using Packet Tracer software.				
3	Exp 3:To Design and implement Static routing using GNS3 software.				
4	Exp 4: To Design and implement Dynamic routing using GNS3 software.				
5	Exp 5: Comparative Analysis of Network Simulators.				
6	Exp 6: Write a program using TCP socket for wired network for following (any one) a. Write Hello to Each other				
7	Exp 7: Write a program using UDP Sockets to enable file transfer between two machines.				
8	Exp 8: Capture packets using Wireshark, write the exact packet capture filter expressions to accomplish the following and save the output in file: a) Capture all TCP traffic to/from Facebook/instagram/youtube, during the time when you log in to your account				
9	Exp 9: To implement and analyze DHCP server and client				
10	Exp 10: To Analyze captured packets to identify potential security issues such as unauthorized access, clear-text credentials, or malicious activity				
11	Assignment 1				
12	Assignment 2				
				TOTAL	

EXPERIMENTS

Expt. No. 1

Title: To Implement and analyze TCP/IP utility commands.

Rubric	***Score***
Delivery(0-3)	
Pre Lab & Post Lab Question Answered (0-3)	
Simulation Output Screenshot Attached With Name & UIN (0-3)	
Total	

Performed On: _____

Sign: _____

Experiment No. 1

Aim : To Implement and analyze TCP/IP utility commands.

Tools Required:

- CMD Command Window

Pre-Lab Questions:

1) What is OS ?

2)What are different types of OS ?

Post-Lab Questions:

1) Name the networking commands for determining the connection of the internet.

2) Explain nslookup command ?

Theory :

1. Ping : The Ping command allows you to test the reachability of a device on a network. Pinging a host should return four data packets, if the data packets are not returned you know there is a problem with your network connection.

To run the basic command, at the prompt type: ping [host]

Where [host] is the name or IP address of a common host server.

2. IPConfig : The IPConfig command is one of the more useful basic Windows network commands everyone should know and use to troubleshoot problems. The IPConfig command displays basic IP address configuration information for Windows.

To run the basic command, at the prompt type: ipconfig

The general information includes IP Addresses for both IPv4 and IPv6, the Default Gateway, and the Subnet Mask. Adding the parameter /all to the command will display DNS Server information and details concerning IP Address leases.

3.Getmac : Every network capable device on the internet has a unique identifying number called its MAC address. The number is assigned during manufacture and is established in the hardware of the device. Using the Getmac command, a user can determine the MAC address of their various network devices. Some administrators will use the unique MAC addresses of devices to limit what can and cannot connect to a network.

To run the basic command, at the prompt type : getmac

4. hostname : The Windows Host Name network command will simply display the current name of your windows computer.

This is the name your computer uses to identify itself to the other devices and servers on your local network. You can find this name in the System information screen in the GUI, but this command is quicker.

To run the basic command, at the prompt type: hostname

5. NSLookUp : The NSLookUp Windows network command displays information that you can use to diagnose Domain Name System (DNS) infrastructure. Using NSLookUp without a parameter will show the DNS server your PC is currently using to resolve domain names into IP addresses.

To run the basic command, at the prompt type: nslookup

6. Tracert : Another handy tool for troubleshooting network connections in Windows is the Tracert command. This command will trace the route a data packet takes before reaching its destination, displaying information on each hop along the route. Each hop of the route will display the latency between your device and that particular hop and the IP address of the hop.

To run the basic command, at the prompt type: tracert [host]

7. Netstat : The Netstat command displays active TCP connections, ports on which the computer is listening, Ethernet statistics, the IP routing table, IPv4 statistics, and IPv6 statistics. When used without parameters, this command displays active TCP connections. The information this command provides can be useful in pinpointing problems in your network connections.

To run the basic command, at the prompt type: netstat

8. Arp : The Windows network command Arp displays entries in the Address Resolution Protocol (ARP) cache, which contains one or more tables that are used to store IP addresses and their resolved Ethernet physical addresses. To get useful information from the Arp command you must provide a parameter. The most general parameter is /a, which displays current Arp cache tables for all interfaces.

To run the basic command, at the prompt type: arp -a

9. PathPing : The Windows network command PathPing combines the ping command with the tracert command, providing information about network latency and network loss at intermediate hops between a source and destination. The PathPing command provides more detail than either ping or tracert can provide, such as latency reports and statistics on packet loss.

To run the basic command, at the prompt type: pathping [host]

Where [host] is the name or IP address of a common host server (google.com, techrepublic.com, etc.)

10. SystemInfo : The last command on our list is the SystemInfo command, which displays a detailed list of configuration information about your Windows PC. The information listed by this command is too lengthy to mention in full but includes the installed version of Windows,

the host name, the Product ID, the type and number of CPUs, RAM configuration, network card details and installed hotfixes.

Output :

Conclusion:

……..
.....……
……..

CO's Covered:

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.....……

.....

Expt. No. 2

Title: To Configure network devices, such as hubs and switches within a network topology using Packet Tracer software.

Rubric	***Score***
Delivery(0-3)	
Pre Lab & Post Lab Question Answered (0-3)	
Simulation Output Screenshot Attached With Name & UIN (0-3)	
Total	

Performed On: _____

Sign: _____

Experiment No. 2

Aim : To Configure network devices, such as hubs and switches within a network topology using Packet Tracer software.

Tools Required:

- **Cisco Packet Tracer**
- **PC/Laptop**
- **Windows Operating System**

Pre-Lab Questions:

1) What is Cisco Packet Tracer ?

Post-Lab Questions:

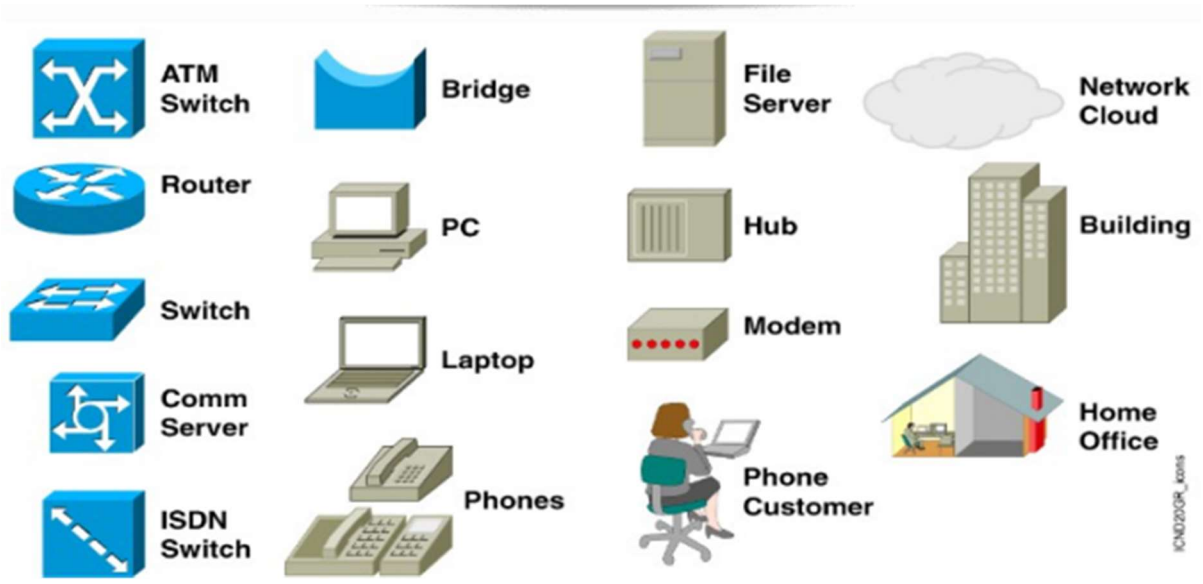
1) Compare Star & Mesh Topology (Min 2 Points)

Theory :

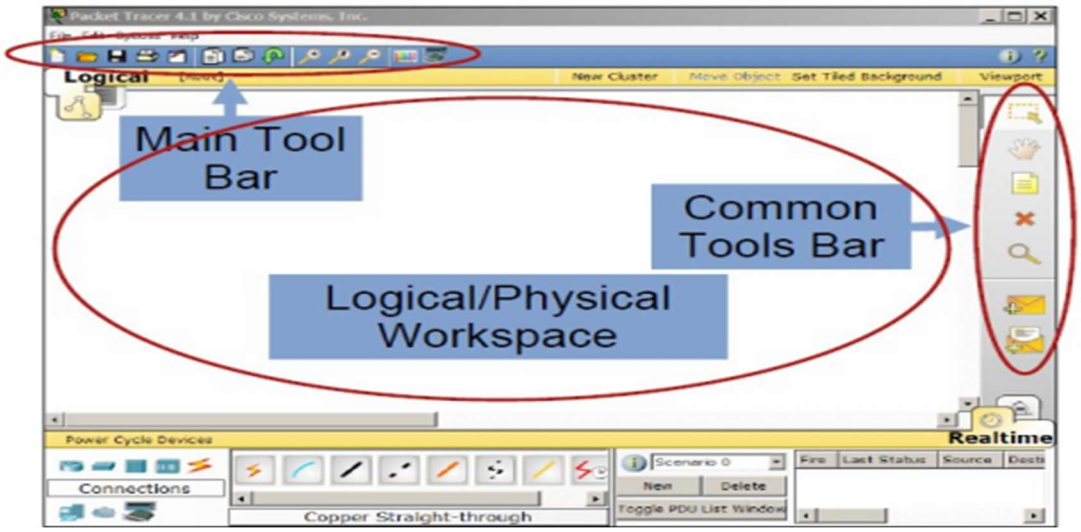
Packet Tracer is a cross-platform visual simulation tool designed by Cisco Systems that allows users to create network topologies and imitate modern computer networks. The software allows users to simulate the configuration of Cisco routers and switches using a simulated command line interface. Packet Tracer makes use of a drag and drop user interface, allowing users to add and remove simulated network devices as they see fit. Packet Tracer allows users to create simulated network topologies by dragging and dropping routers, switches and various other types of network devices. A physical connection between devices is represented by a "cable" item. Packet Tracer supports a multi-user system that enables multiple users to connect multiple topologies together over a computer network. The software is mainly focused on Certified Cisco Network Associate Academy students as an educational tool for helping them learn fundamental CCNA concepts.

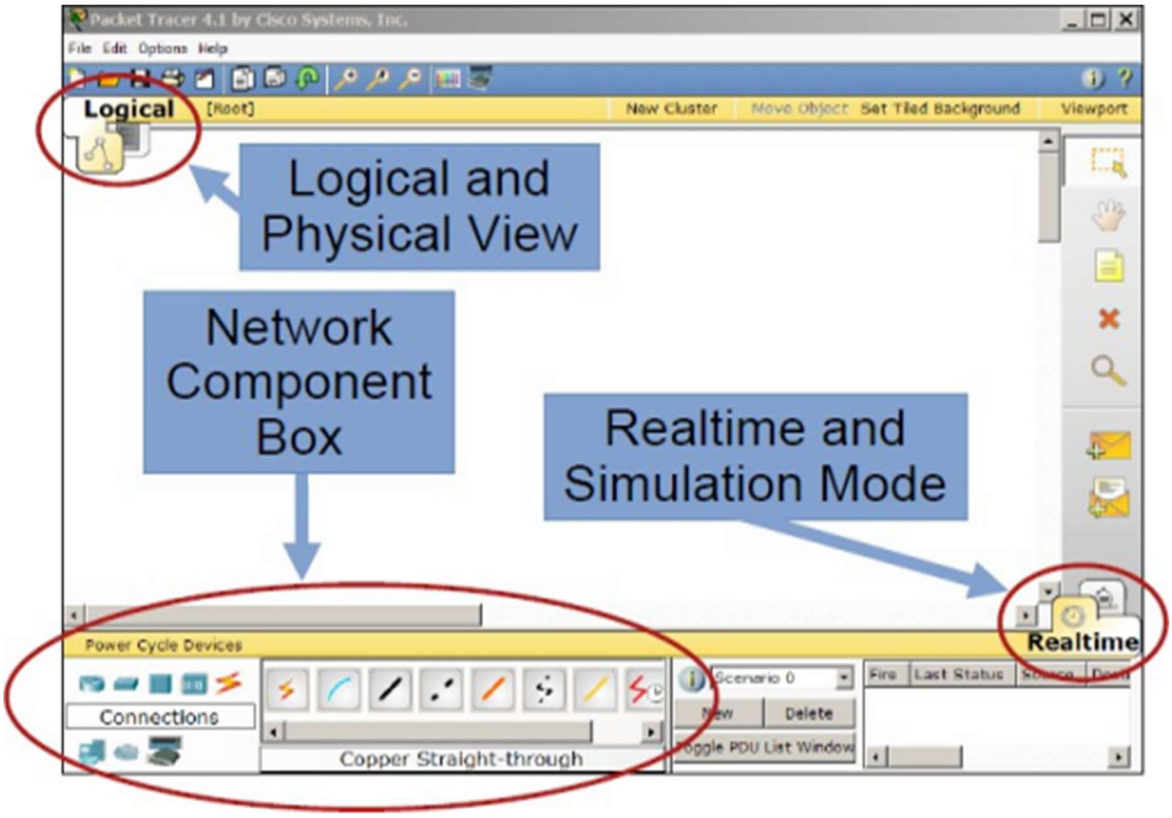
Following devices to be studied

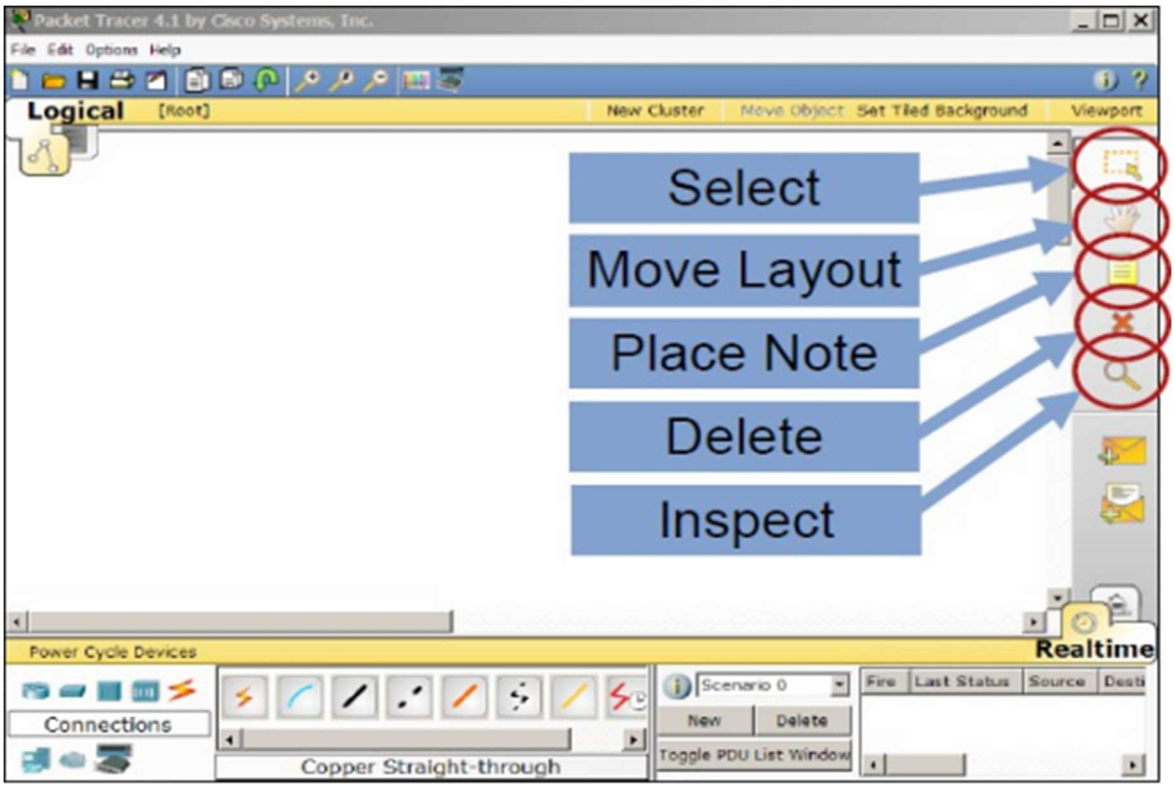
· **Study of network devices**

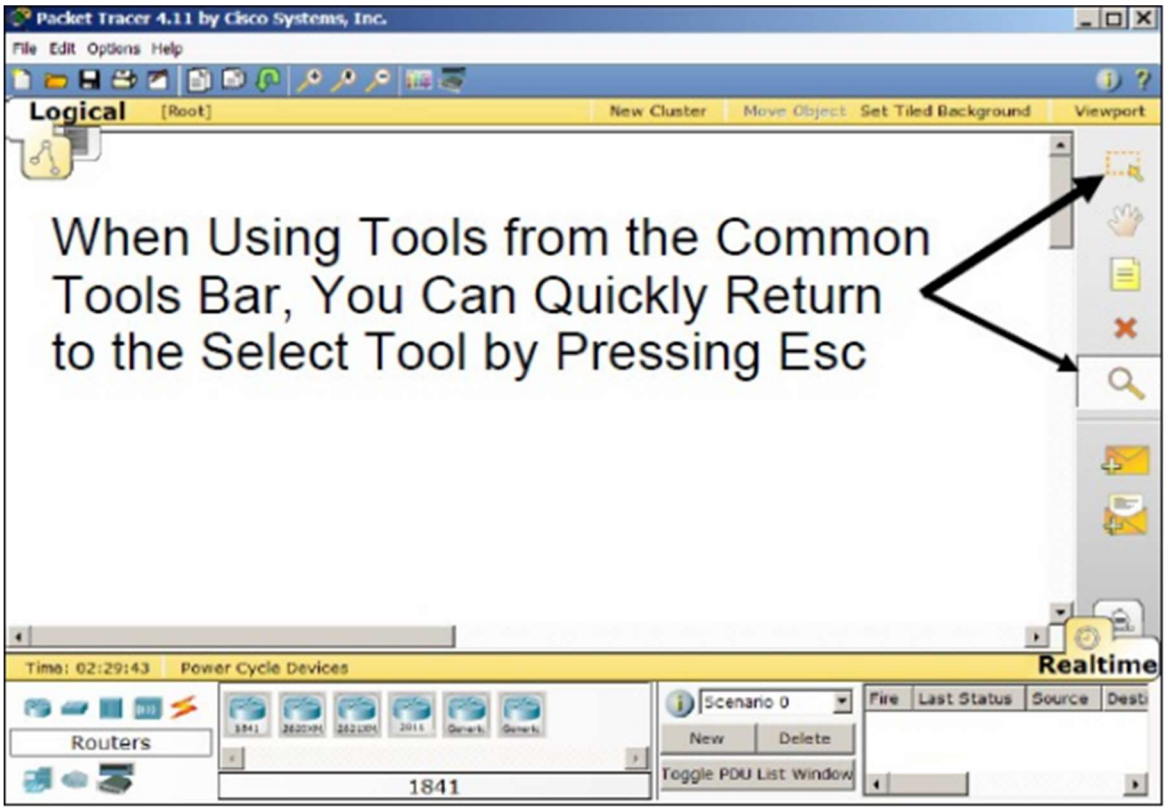


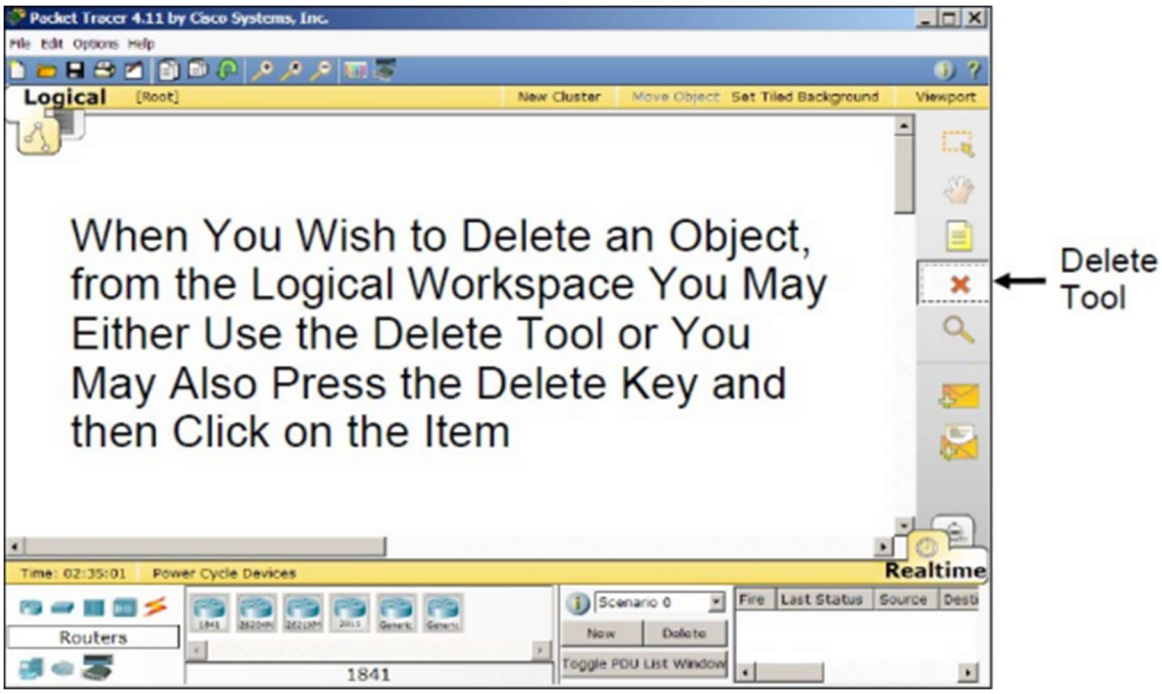
Overview of the Packet Tracer Interface

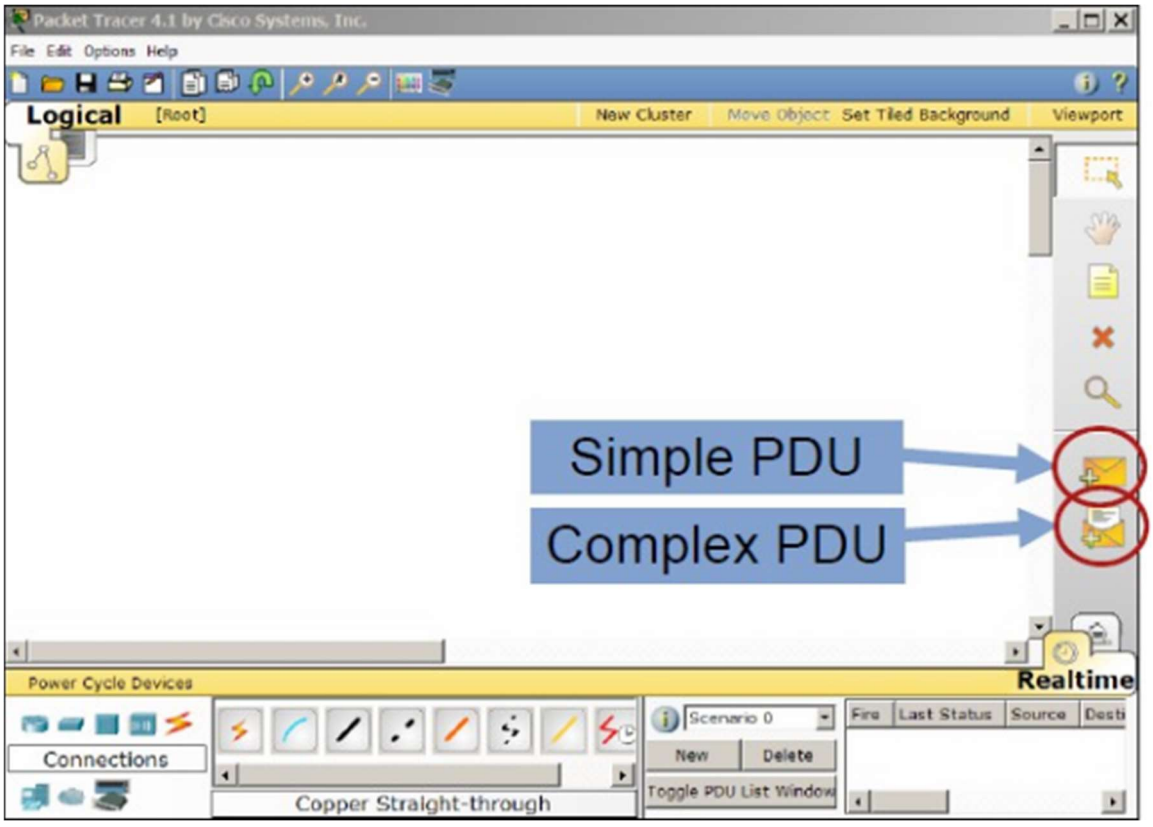





Packet Tracer 4.1 by Cisco Systems, Inc.
File Edit Options Help
Logical [Root]
New Cluster Move Object Set Tiled Background Viewport
Select
Move Layout
Place Note
Delete
Inspect
Power Cycle Devices
Realtime
Connections
Copper Straight-through
Scenario 0
New Delete
Toggle PDU List Window
Fire Last Status Source Desti

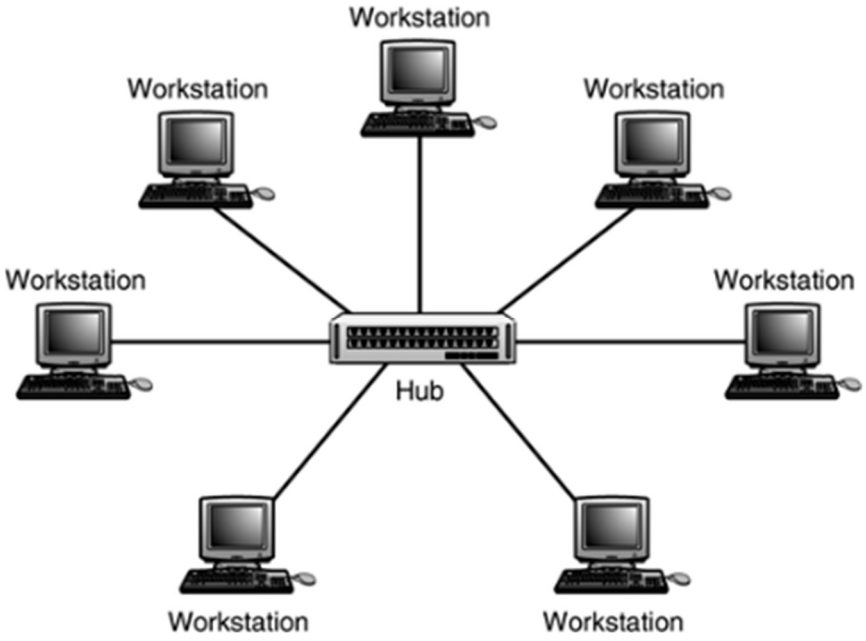

Packet Tracer 4.11 by Cisco Systems, Inc.
File Edit Options Help
Logical [Root]
New Cluster Move Object Set Tiled Background Viewport
When Using Tools from the Common Tools Bar, You Can Quickly Return to the Select Tool by Pressing Esc
Time: 02:29:43 Power Cycle Devices
Realtime
Routers
1841
Scenario 0
New Delete
Toggle PDU List Window
Fire Last Status Source Desti


When You Wish to Delete an Object, from the Logical Workspace You May Either Use the Delete Tool or You May Also Press the Delete Key and then Click on the Item
Delete Tool
Logical
Realtime
Routers
1841


Simple PDU
Complex PDU
Logical
Realtime
Power Cycle Devices
Connections
Copper Straight-through

Implementation of STAR Topology in Cisco Packet Tracer :

In Star topology, all the components of a network are connected to the central device called "hub" which may be a router or a switch. All the workstations are connected to a central device with a point-to-point connection. So, it can be said that every computer is indirectly connected to every other node by the help of "hub". All the data on the star topology passes through the central device before reaching the intended destination. Hub acts as a junction to connect different nodes present in Star Network, and at the same time it manages and controls the whole of the network. Depending on which central device is used, "hub" can act as repeater or signal booster. Central devices can also communicate with other hubs of different networks. Unshielded Twisted Pair (UTP) Ethernet cable is used to connect workstations to central node.



Advantages of Star Topology:

1) A sent signal reaches the intended destination after passing through not more than 3-4 devices and 2-3 links. Performance of the network is dependent on the capacity of the central hub.

2) Easy to connect new nodes or devices. In star topology new nodes can be added easily without affecting the rest of the network. Similarly, components can also be removed easily.

3) Centralized management. It helps in monitoring the network.

4) Failure of one node or link doesn't affect the rest of the network. At the same time it's easy to detect the failure and troubleshoot it.

Disadvantages of Star Topology:

1) Too much dependency on the central device. If it fails the whole network fails.

2) The use of a hub, a router or a switch as a central device increases the overall cost of the network.

3) Performance and as well number of nodes which can be added in such topology is dependent on the capacity of the central device.

Procedure:

1. Open Cisco Packet Tracer.

2. Drag and drop the required components required for the Star Topology.

3. Connect all the nodes and the central co-ordinator.

4. Apply the IP address, Subnet Mask and Default Gateway.

5. Add simple PDU at the various nodes.

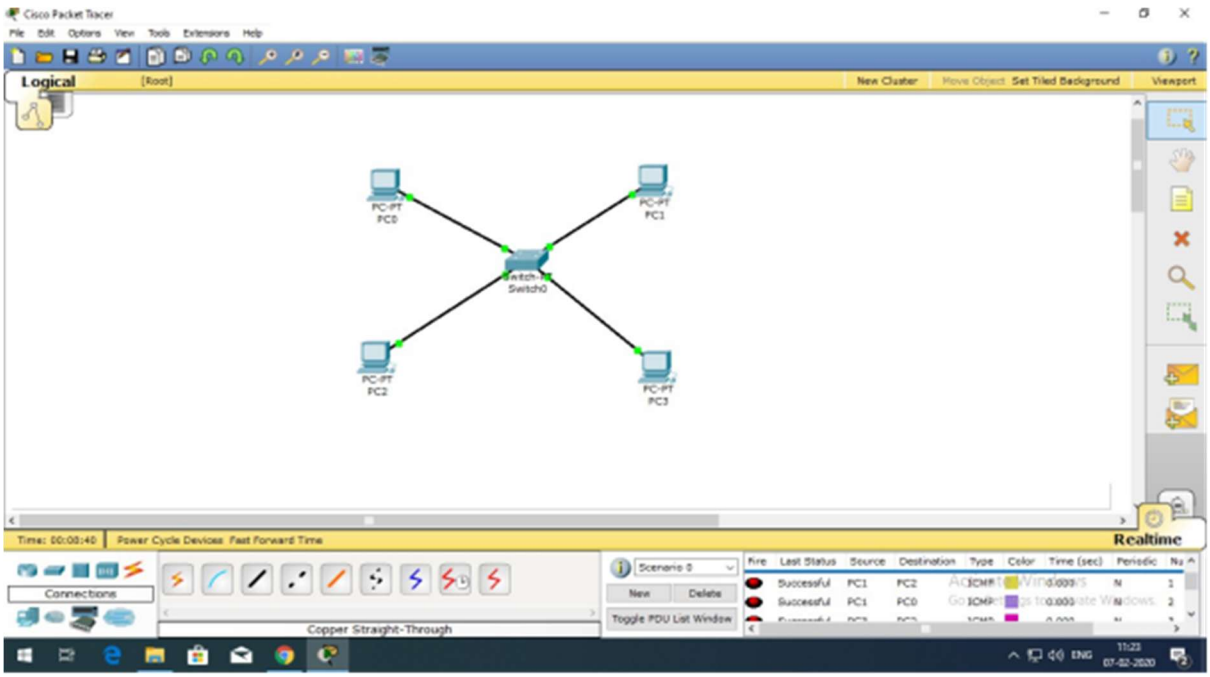
6. Take a snapshot of the configuration and paste in a word page under the title 'Scenario 1 Star topology before Simulation'

7. Run the simulation and observe the flow of data packets.

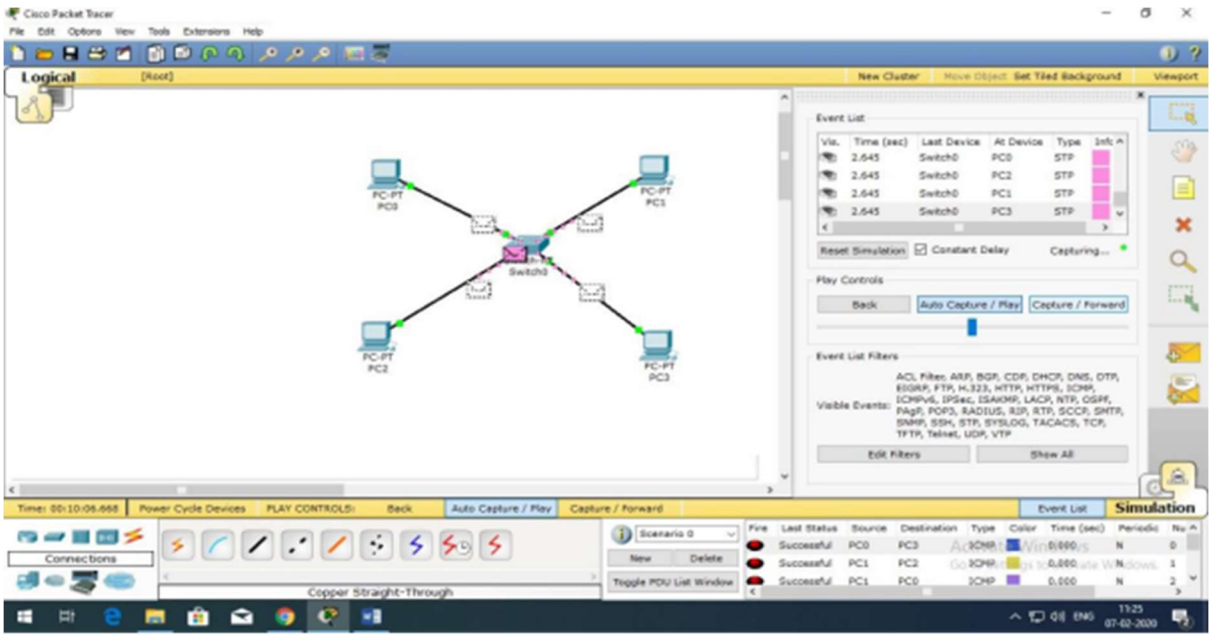
8. Again, take a snapshot and on a word page below the Scenario 1 and label it 'Scenario 2 Star Topology after Simulation'.

Design a star topology **using a switch as a central co-ordinator and four nodes.**

Scenario 1: Star topology before simulation :



Scenario 2: Star topology after simulation


Cisco Packet Tracer
Logical
PC-PT PC0
PC-PT PC1
PC-PT PC2
PC-PT PC3
Switch0
Event List
Play Controls
Back
Auto Capture / Play
Capture / Forward
Event List Filters
Edit Filters
Show All
Simulation
Connections
Copper Straight-Through

Output:

Conclusion:

……..
……
……..

CO's Covered:

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Expt. No.3

Title: To Design and implement Static routing using GNS3 software.

Rubric	***Score***
Delivery(0-3)	
Pre Lab & Post Lab Question Answered (0-3)	
Simulation Output Screenshot Attached With Name & Uin (0-3)	
Total	

Performed On: _____

Sign: _____

Experiment No. 3

Aim : To Design and implement Static routing using GNS3 software.

Tools Required:

- GNS3
- PC/Laptop
- Windows Operating System

Pre-Lab Questions:

1) What is the function of Network Layer ?

2)What is Routing ?

Post-Lab Questions:

1) What is Static Routing & its advantages ?

2) List the Disadvantages of Static Routing ?

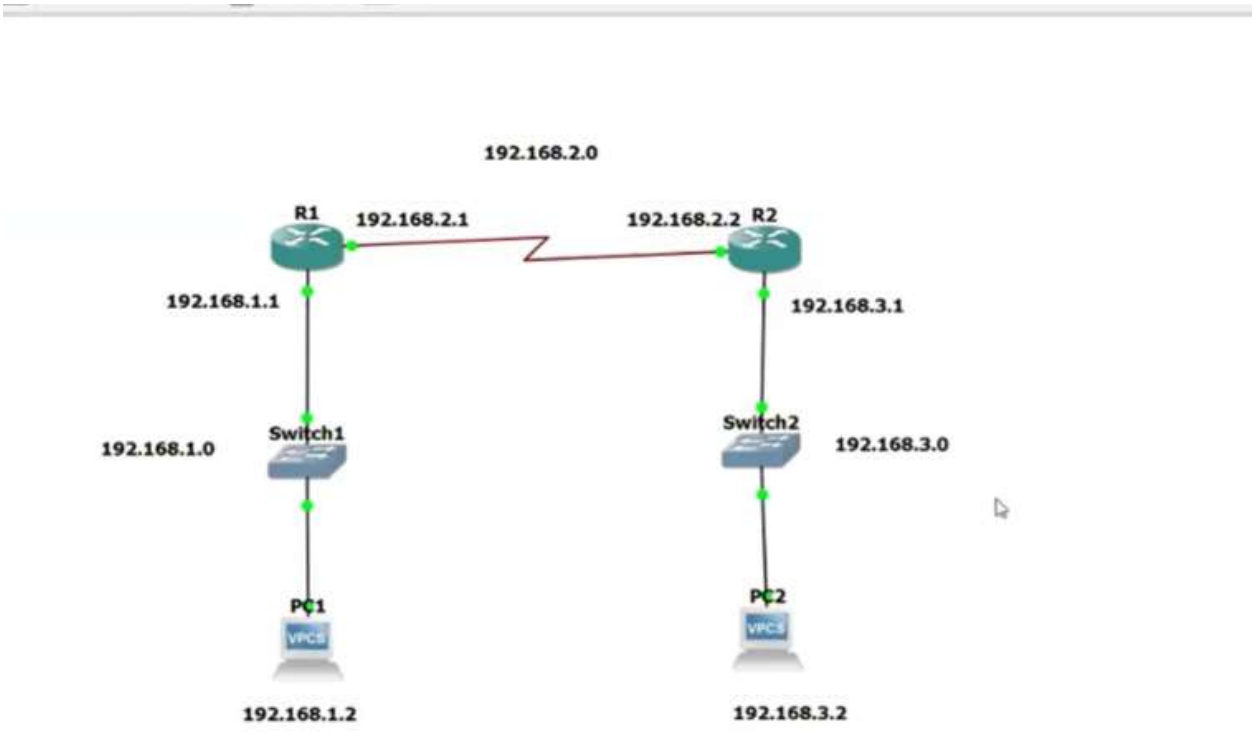
Theory

Static routing is a method used in computer networking to manually configure the routing table of a network device, such as a router or a switch, to determine how data packets should be forwarded within a network. In static routing, network administrators define specific routes, indicating the next hop or next router to which a packet should be sent based on its destination IP address.

Why Use Static Routing?

- Simple networks with few routers
- Single path from source to destination
- Security (no routing protocol traffic)
- Bandwidth conservation
- Predictable behavior

Network Diagram :



Procedure :

Phase 1: Build the Topology in GNS3

1. Launch GNS3 and create a new project (e.g., Static_Routing_Lab)
2. Drag and drop the following devices onto the workspace:
    - 2 × Cisco Routers (e.g., c2600 or c7200 IOS)
    - 2 × Ethernet Switches
    - 4 × VPCS (or end devices)
3. Connect cables:
    - PC1 → SW1 (Ethernet)
    - PC2 → SW1 (Ethernet)
    - SW1 → R1 Fa0/0 (Ethernet)
    - PC3 → SW2 (Ethernet)
    - PC4 → SW2 (Ethernet)
    - SW2 → R2 Fa0/0 (Ethernet)
    - R1 Fa0/1 → R2 Fa0/1 (Serial or Crossover Ethernet)
4. Start all devices (green play button)

5. Configure R1 & R2 & End Devices

Output:

Conclusion:

……..
……
……..

CO's Covered:

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……..

Expt. No. 4

Title: Design And Implement Dynamic Routing Using Cisco Packet Tracer

Rubric	***Score***
Delivery(0-3)	
Pre Lab & Post Lab Question Answered (0-3)	
Simulation Output Screenshot Attached With Name & Uin (0-3)	
Total	

Performed On: _____

Sign: _____

Experiment No. 4

Aim : To Design And Implement Dynamic Routing Using Cisco Packet Tracer.

Tools Required:

- Cisco Packet Tracer
- PC/Laptop
- Windows Operating System

Pre-Lab Questions:

1) What is dynamic routing, and how is it different from static routing?

2) What is RIP, and what metric does it use to determine the best path?

Post-Lab Questions:

1) How did you verify that dynamic routing was successfully configured in Cisco Packet Tracer?

Theory :

Dynamic Routing is a routing technique in which routers automatically exchange routing information with each other using routing protocols. Unlike Static Routing, routes are updated automatically whenever the network topology changes.

Dynamic routing protocols reduce the administrator's workload and provide better scalability for large networks.

Advantages

- Automatic route updates.
- Easy to manage large networks.
- Supports network scalability.
- Provides alternate routes in case of link failure.

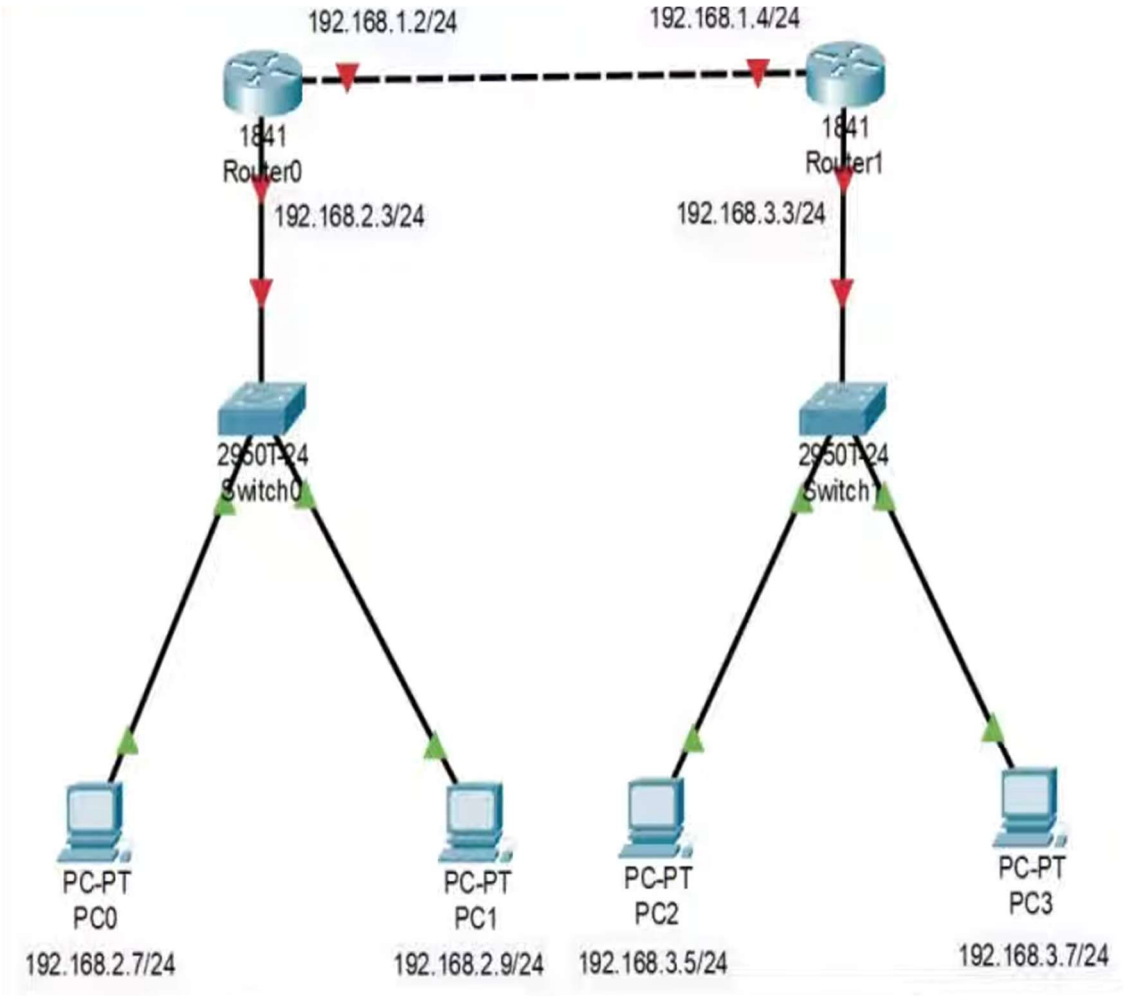
Disadvantages

- Uses CPU and bandwidth.
- Slightly slower than static routing.
- Requires routing protocol configuration**.**

Common Dynamic Routing Protocols:

- RIP (Routing Information Protocol)
- OSPF (Open Shortest Path First)
- EIGRP (Enhanced Interior Gateway Routing Protocol)

NETWORK DIAGRAM:



Output:

Conclusion:

……..... ..... …… ……..

CO'sCovered:

……..... ..... …… ……..

Expt. No. 5

Title: Comparative Analysis of Network Simulators.

Rubric	***Score***
Delivery(0-3)	
Pre Lab & Post Lab Question Answered (0-3)	
Simulation Output Screenshot Attached With Name & UIN (0-3)	
Total	

Performed On: _____

Sign: _____

Experiment No. 5

Aim : Comparative Analysis of Network Simulators using Star Topology

Tools Required:

- Cisco Packet Tracer & GNS3
- PC/Laptop
- Windows Operating System

Pre-Lab Questions:

1) What is a star topology? Explain its advantages and disadvantages in a computer network.

2) Differentiate between Cisco Packet Tracer and GNS3 based on their features, simulation capabilities, and practical applications.

Post-Lab Questions:

1) **After implementing the same Star Topology in Cisco Packet Tracer & GNS3 compare the three network simulators in terms of ease of configuration, simulation accuracy, resource utilization, visualization, and suitability for learning, enterprise networking, and research. Which simulator would you recommend for each application and why?**

Theory :

Comparative Analysis of Network Simulators Using Star Topology involves evaluating the performance, features, and usability of different network simulation tools by implementing the same star topology in each simulator. In a **Star Topology**, all end devices are connected to a central network device (usually a switch or hub), making it one of the most widely used topologies in Local Area Networks (LANs).

Cisco Packet Tracer is a network simulation tool developed by Cisco that provides a user-friendly graphical interface for designing, configuring, and troubleshooting Cisco-based networks. It is primarily used for educational purposes and CCNA/CCNP training.

GNS3 (Graphical Network Simulator-3) is an advanced network emulation platform that allows users to run real Cisco IOS images and other network operating systems. It provides a realistic networking environment suitable for research, testing, and enterprise-level network design.

Advantages

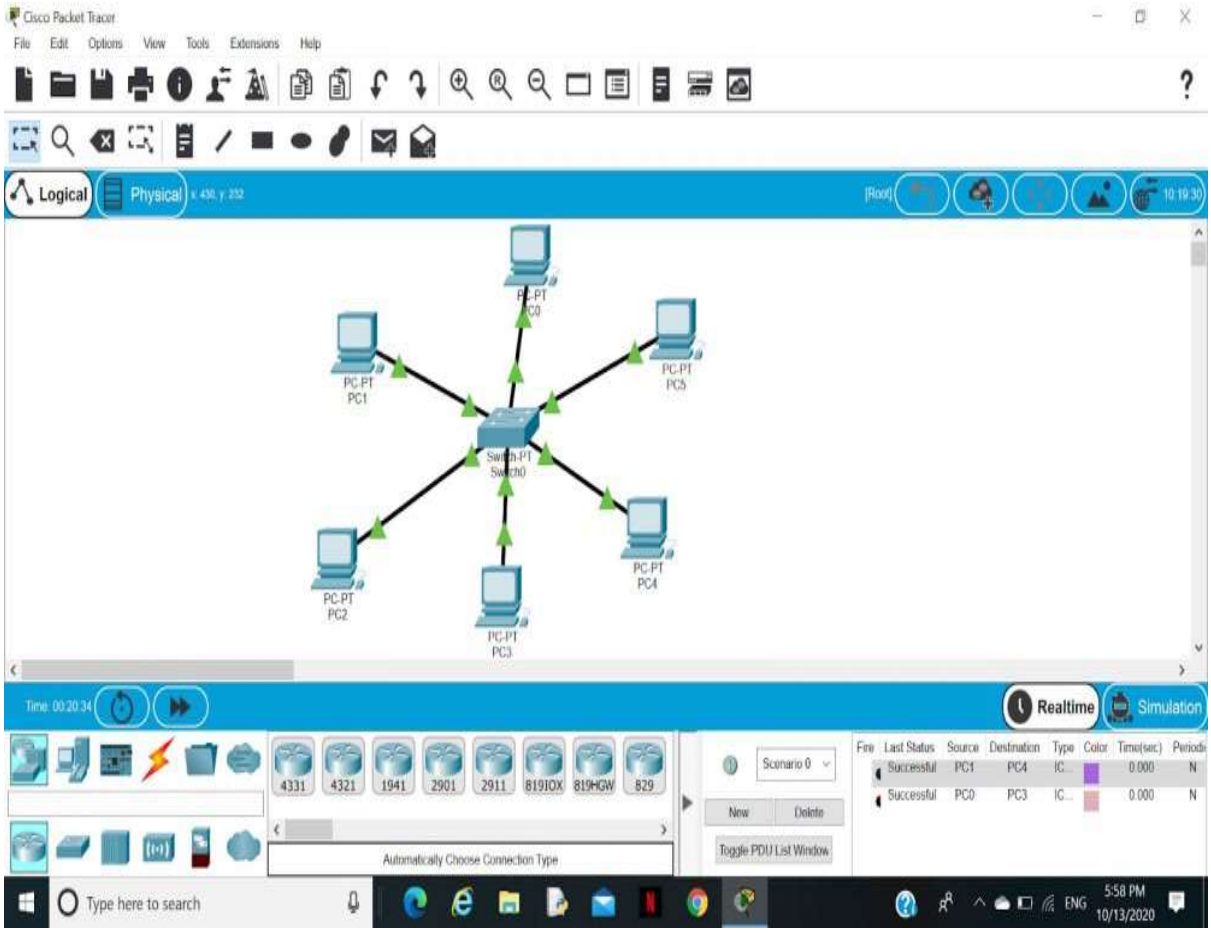
- Easy visualization and analysis of network topology.
- Enables testing without physical networking hardware.
- Cost-effective and safe environment for experiments.
- Helps compare different simulators based on performance and features.
- Useful for learning, research, and network troubleshooting.

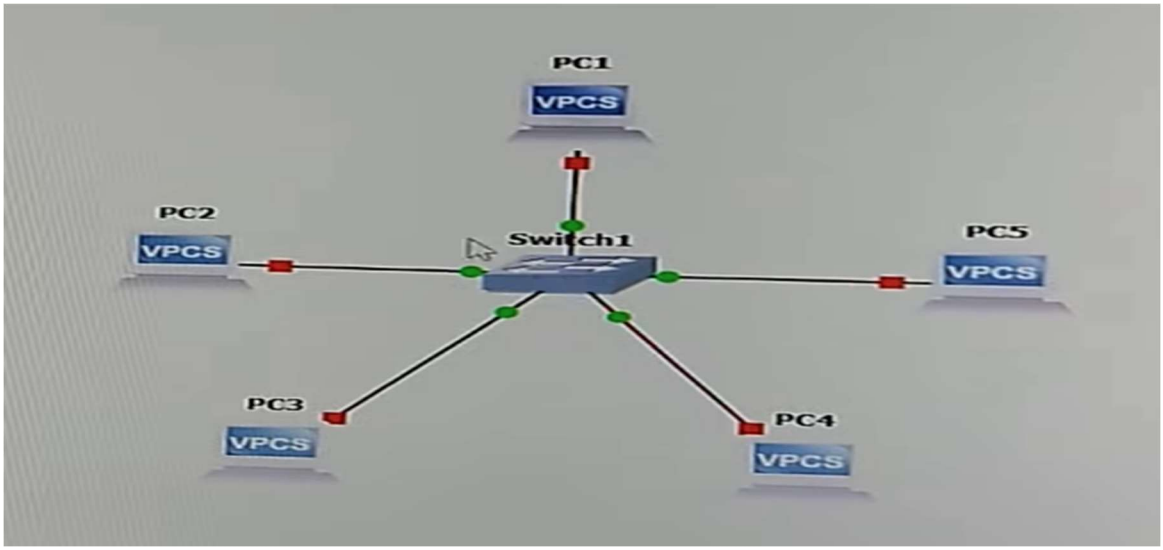
Disadvantages

- Simulation results may differ from real-world network performance.
- GNS3 requires higher system resources than Packet Tracer.

- Cisco Packet Tracer supports only Cisco-specific devices and features.
- GNS3 setup is more complex and requires additional configurations.

NETWORK DIAGRAM:





Output:

Conclusion:

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CO's Covered:

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Expt. No. 6

Title: Write a program using TCP socket for wired network for following

a. Write Hello to Each other

Rubric	***Score***
Delivery(0-3)	
Pre Lab & Post Lab Question Answered (0-3)	
Simulation Output Screenshot Attached With Name & UIN (0-3)	
Total	

Performed On: _____

Sign:

Experiment No. 6

Aim : Write a program using TCP socket for wired network for following

a) Write Hello to Each other

Tools Required:

- Operating System: Windows/Linux/macOS
- PC/Laptop
- Python 3.x
- Python built-in socket module
- VS Code/Py-Charm/IDLE (or any Python editor)
- Command Prompt/Terminal
-

Pre-Lab Questions:

1) What is TCP, and why is it considered a connection-oriented protocol?

2) What are the functions of the bind(), listen(), and accept() methods in a TCP server program?

Post-Lab Questions:

1) What would happen if the client attempts to connect before the server starts listening, and how can this issue be resolved ?

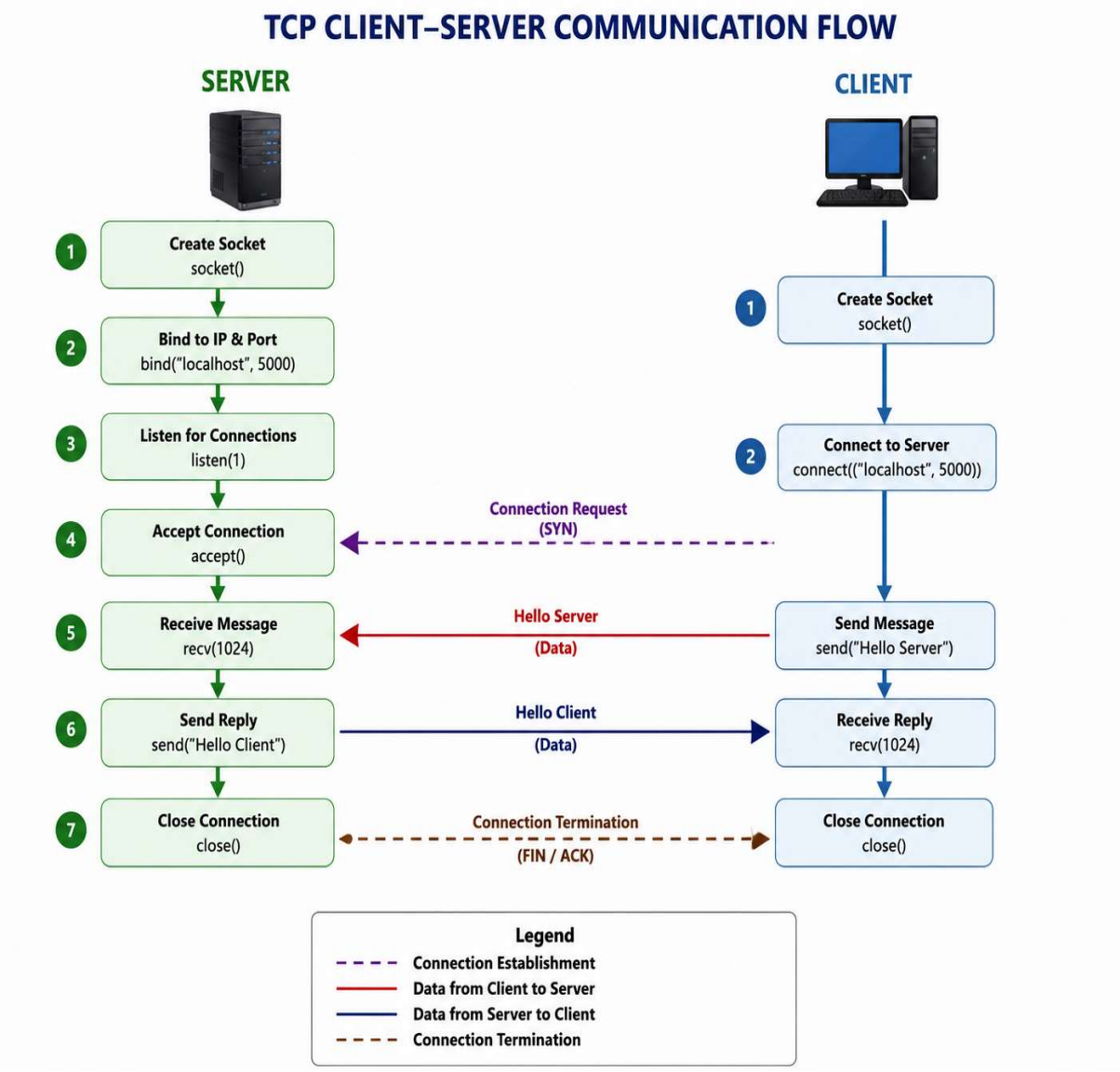
Theory

TCP (Transmission Control Protocol) is a connection-oriented protocol that provides reliable communication between two computers.

A TCP communication involves:

- Creating a socket
- Binding it to an IP address and port (server)
- Listening for client connections
- Accepting the connection
- Sending and receiving messages
- Closing the connection

Communication Flow:



Output:

Conclusion:

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CO's Covered:

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Expt. No. 7

Title: Write a program using UDP Sockets to enable file transfer between two machines.

Rubric	***Score***
Delivery(0-3)	
Pre Lab & Post Lab Question Answered (0-3)	
Simulation Output Screenshot Attached With Name & Uin (0-3)	
Total	

Performed On: _____

Sign: _____

Experiment No. 7

Aim : Write a program using UDP Sockets to enable file transfer between two machines.

Tools Required:

- Python 3.x
- Visual Studio Code (VS Code) / PyCharm / IDLE (Python IDE)
- Command Prompt (Windows) or Terminal (Linux/macOS)
- Socket Library (built-in Python module, no installation required)
- Wireshark (optional, for monitoring and analyzing UDP packets)
- Operating System: Windows/Linux/macOS

Pre-Lab Questions:

1) What is UDP (User Datagram Protocol)? How does it differ from TCP in terms of connection establishment, reliability, and data transmission?

2) Explain the functions of the sendto() and recvfrom() methods in UDP socket programming. Why is an EOF (End of File) marker used during file transfer?

Post-Lab Questions:

1) After implementing file transfer using UDP sockets, what are the limitations of UDP for file transmission, and how can reliability be improved in UDP-based applications?

Theory :
UDP (User Datagram Protocol) is a connectionless protocol. Unlike TCP, UDP does not establish a connection before sending data. Data is transmitted in the form of packets (data-grams).

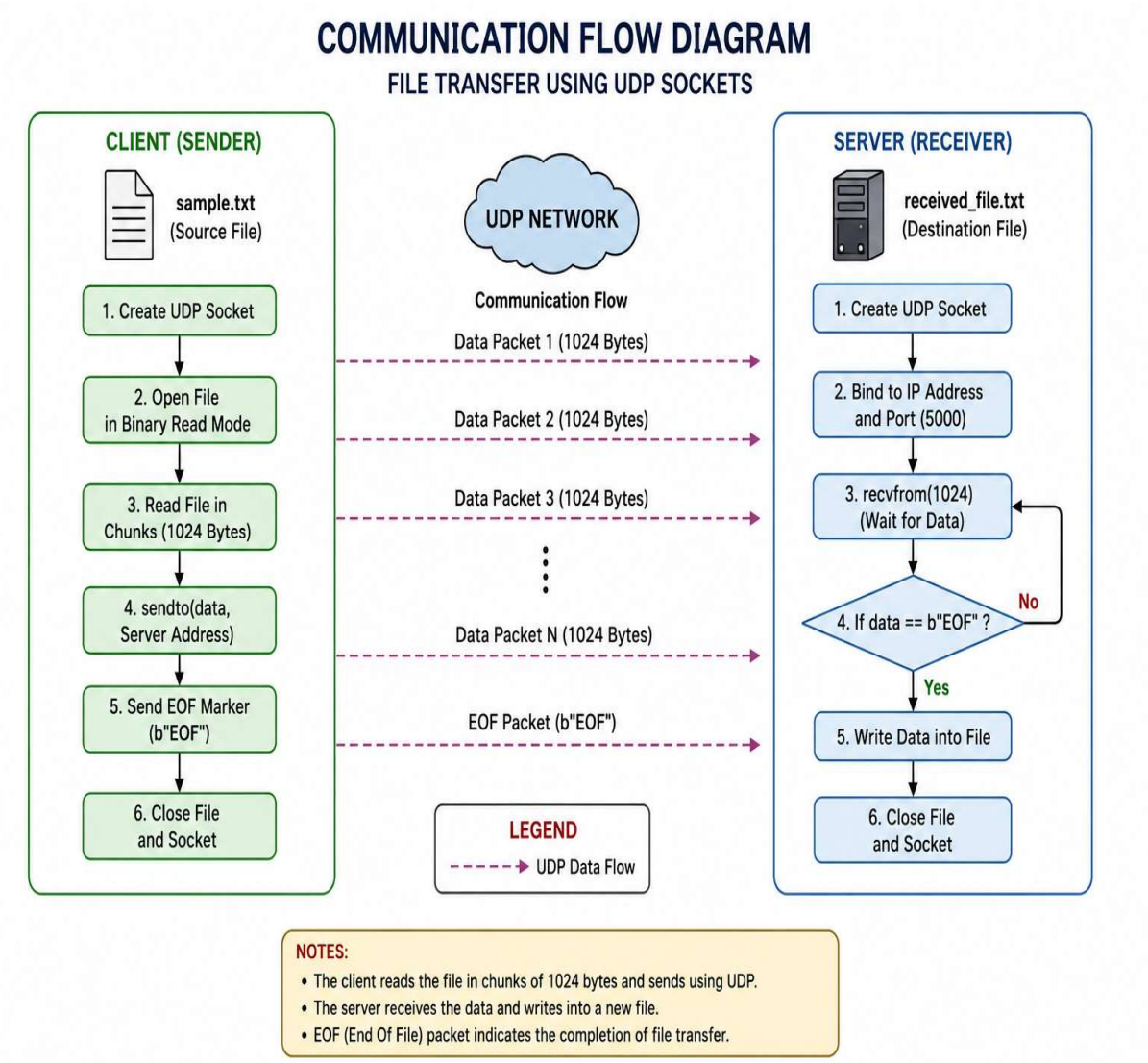
Characteristics of UDP

- Connectionless communication
- Faster than TCP
- No guarantee of delivery
- No packet ordering
- Lower overhead

In this experiment:

- The Client reads a file and sends its contents to the server.
- The Server receives the file data and stores it in a new file.

Communication Flow:


COMMUNICATION FLOW DIAGRAM
FILE TRANSFER USING UDP SOCKETS
CLIENT (SENDER)
sample.txt
(Source File)
1. Create UDP Socket
2. Open File in Binary Read Mode
3. Read File in Chunks (1024 Bytes)
4. sendto(data, Server Address)
5. Send EOF Marker (b"EOF")
6. Close File and Socket
UDP NETWORK
Communication Flow
Data Packet 1 (1024 Bytes)
Data Packet 2 (1024 Bytes)
Data Packet 3 (1024 Bytes)
Data Packet N (1024 Bytes)
EOF Packet (b"EOF")
LEGEND
UDP Data Flow
SERVER (RECEIVER)
received_file.txt
(Destination File)
1. Create UDP Socket
2. Bind to IP Address and Port (5000)
3. recvfrom(1024) (Wait for Data)
4. If data == b"EOF" ?
No
Yes
5. Write Data into File
6. Close File and Socket
NOTES:
• The client reads the file in chunks of 1024 bytes and sends using UDP.
• The server receives the data and writes into a new file.
• EOF (End Of File) packet indicates the completion of file transfer.

Output:

Conclusion:

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……..

CO's Covered:

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Expt. No. 8

Title: To Capture packets using Wireshark, write the exact packet capture filter expressions to accomplish the following and save the output in file:
a) Capture all TCP traffic to/from Facebook/instagram/youtube, during the time when you log in to your account.

Rubric	***Score***
Delivery(0-3)	
Pre Lab & Post Lab Question Answered (0-3)	
Simulation Output Screenshot Attached With Name & Uin (0-3)	
Total	

Performed On: _____

Sign: _____

Experiment No. 8

Aim : To Capture packets using Wireshark, write the exact packet capture filter expressions to accomplish the following and save the output in file:

a) Capture all TCP traffic to/from Facebook/instagram/youtube, during the time when you log in to your account.

Tools Required:

Windows 10/11 or Linux

Wireshark (Latest Version)

Pre-Lab Questions:

1) What is the difference between a Capture Filter and a Display Filter in Wireshark? Explain with one example of each.

Post-Lab Questions:

1) Which protocols were observed while logging into Facebook, Instagram, or YouTube? Explain the role of TCP, DNS, and TLS in the communication process.

2) List the applications of Wireshark ?

Theory :

Wireshark is an open-source network protocol analyzer used to capture, monitor, and analyze packets transmitted over a computer network.

It captures packets in real time and provides detailed information about:

- Source IP Address
- Destination IP Address
- Protocol
- Port Numbers
- Packet Length
- TCP Flags
- TLS/SSL Handshake
- HTTP/HTTPS Requests
- DNS Queries

It is widely used for:

- Network Troubleshooting
- Cyber Security
- Protocol Analysis
- Performance Monitoring

- Network Forensics

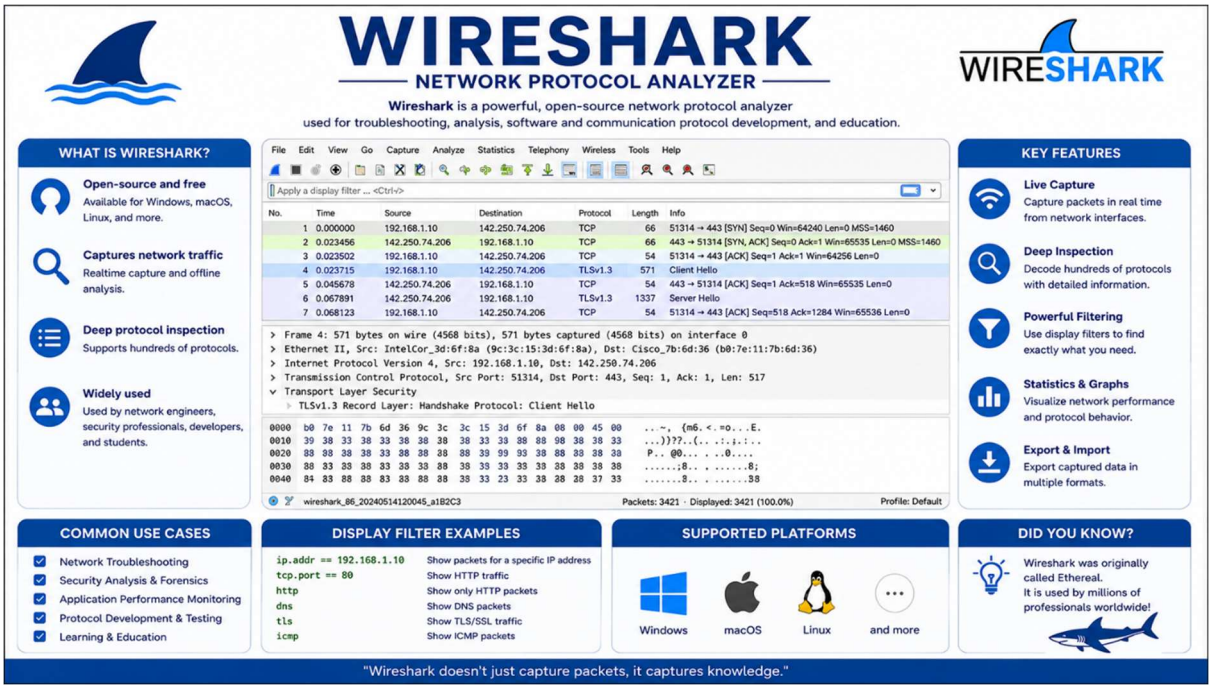


Fig 1 : Wireshark

Procedure :

Step 1
Install Wireshark.

Step 2

Launch Wireshark as Administrator.

Step 3

Select the active network interface.

Example

- Wi-Fi
- Ethernet

Step 4

In the **Capture Filter** box enter

tcp

Step 5

Click

Start Capturing Packets

(or double-click the network interface).

Step 6

Open Google Chrome.

Visit

https://www.facebook.com

Login using your credentials.

Wait for

- Home page to load
- Notifications
- Images

Step 7

Repeat for

https://www.instagram.com

Login successfully.

Step 8

Repeat for

https://www.youtube.com

Login into YouTube.

Step 9

Return to Wireshark.

Click

Stop Capture

Step 10

Save Capture

File

↓

Save As

Example

facebook_login.pcapng

or

instagram_login.pcapng

or

youtube_login.pcapng

Analyze Packets

Apply Display Filter

tcp

Observe

- Source IP
- Destination IP
- Source Port
- Destination Port
- TCP Flags
- Packet Length
- Sequence Number
- ACK Number

Output:

Conclusion:

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.....……

CO's Covered:

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Expt. No. 9

Title: To implement and analyze DHCP server and client

Rubric	*Score*
Delivery(0-3)	
Pre Lab & Post Lab Question Answered (0-3)	
Simulation Output Screenshot Attached With Name & UIN (0-3)	
Total	

Performed On: _____

Sign: _____

Experiment No. 9

Aim: To implement and analyze DHCP server and client

Tools Required:

- Operating System: Windows/Linux/macOS
- PC/Laptop
- Python 3.x
- Python built-in socket module
- VS Code/PyCharm/IDLE (or any Python editor)
- Command Prompt/Terminal

Pre-Lab Questions:

1) What is DHCP ?

2) What is DORA ?

Post-Lab Questions:

1) Explain the Flow between DHCP Client & Server ?

Theory

DHCP (Dynamic Host Configuration Protocol) is a client-server network protocol that automatically assigns IP addresses and other network configuration parameters (subnet mask, default gateway, DNS server) to devices on a network. It operates on UDP ports 67 (server) and 68 (client).

Without DHCP, every device would need to be manually configured with a static IP address, which is impractical in large networks.

The DORA Process (How DHCP Works)

When a client device boots up and requests an IP address, DHCP follows a 4-step handshake known as DORA:

Step 1: DHCP Discover

- The client broadcasts a message on the network saying *"I need an IP address."*
- Since the client has no IP yet, it uses 0.0.0.0 as source and 255.255.255.255 as destination.
- This is a broadcast — every device on the local subnet hears it.

Step 2: DHCP Offer

- The DHCP server receives the Discover message and responds with an Offer.
- The offer contains a proposed IP address, subnet mask, lease duration, and other options.
- The server marks this IP as "reserved" for that client.

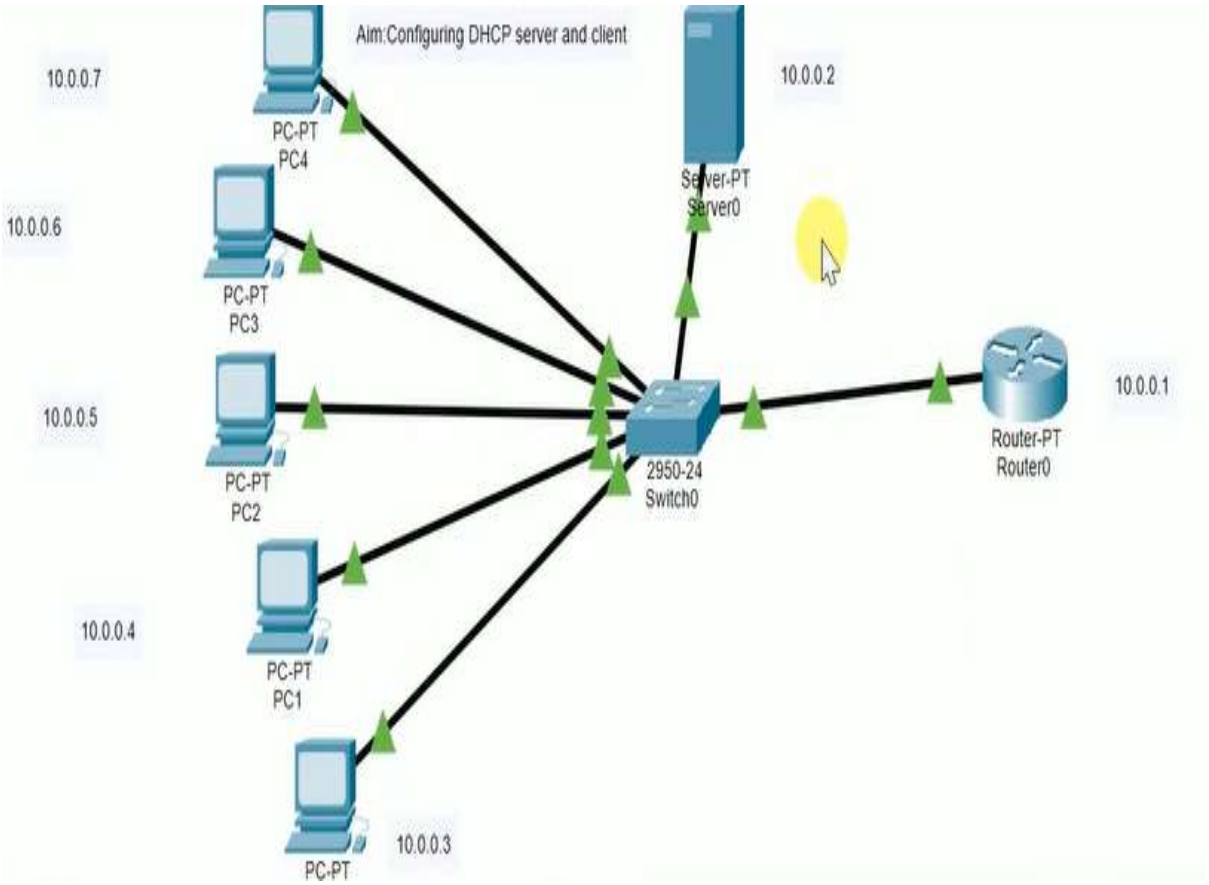
Step 3: DHCP Request

- The client receives one or more offers and broadcasts a Request to accept one.
- It explicitly tells the chosen server: *"I accept your offer."*
- It also implicitly tells other servers: *"I decline your offers."*

Step 4: DHCP ACK (Acknowledgment)

- The server sends a final ACK confirming the lease.
- The client now configures its network interface with the assigned IP and parameters.

Network Diagram:



Procedure :

Step-by-Step Guide

Step 1: Build the Network Topology

Place the following devices in Packet Tracer:

- 1 Router (e.g., 2911 or generic PT Router)
- 1 Switch (or 2 switches for multiple subnets)
- 1 Server-PT (this will be your DHCP Server)
- Multiple PCs (DHCP clients)

Connect them using Copper Straight-Through cables:

- Router ↔ Switch
- Switch ↔ PCs
- Router ↔ DHCP Server (on a separate interface)

Step 2: Configure the DHCP Server (Static IP)

1. Click on the Server-PT device.
2. Go to Desktop → IP Configuration.
3. Set a static IP address for the server (do NOT use DHCP here):

Setting	Value
IPv4 Address	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1

The Default Gateway is the Router interface IP that faces the server.

Step 3: Enable & Configure DHCP Service on the Server

1. Click on the Server-PT → go to Services tab.
2. Click DHCP from the left menu.
3. Turn ON the DHCP service.
4. Delete any default pool values.
5. Create a new DHCP pool. Fill in:

Table	
Pool Name	Pool1 (or any name)
Default Gateway	192.168.10.1
DNS Server	8.8.8.8 (optional)
Start IP Address	192.168.10.3
Subnet Mask	255.255.255.0
Maximum Users	20 (or your preferred range)

Click Save.

6. If you have a second subnet, create another pool (e.g., Pool2 with gateway 192.168.20.1 and start IP 192.168.20.2).

Step 4: Configure the Router

Open the Router's CLI and enter the following commands:

```
enable
configure terminal

! Configure interface facing the DHCP Server
interface GigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.0
 no shutdown
 exit

! Configure interface facing the first subnet (Clients)
interface GigabitEthernet0/1
 ip address 192.168.20.1 255.255.255.0
 ip helper-address 192.168.10.2   ! Points to DHCP Server
 no shutdown
 exit

! Save configuration
do write memory
```

The ip helper-address 192.168.10.2 command is critical — it tells the router to forward DHCP broadcast requests from the client subnet to the DHCP server.

Step 5: Configure PCs to Use DHCP

1. Click on any PC.
2. Go to Desktop → IP Configuration.
3. Change the setting from Static to DHCP.
4. The PC will automatically request and receive an IP address from the DHCP server.
5. You should see it get an IP like 192.168.20.2 (or from the appropriate pool).

Repeat for all PCs

Step 6: Verify the Configuration

On any PC, open the Command Prompt and type:

```
bash
```

ipconfig

You should see:

- IPv4 Address (dynamically assigned)
- Subnet Mask
- Default Gateway

To verify on the Router:

```
bash
```

show ip dhcp binding

Output:

Conclusion:

……..
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……..

CO's Covered:

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……..

Expt. No. 10

Title: To analyze captured network packets using Wireshark to identify potential security issues such as unauthorized access attempts, clear-text credentials, suspicious communication, and other anomalous network activities.

Rubric	***Score***
Delivery(0-3)	
Pre Lab & Post Lab Question Answered (0-3)	
Simulation Output Screenshot Attached With Name & Uin (0-3)	
Total	

Performed On: _____

Sign: _____

Experiment No. 10

Aim : To analyze captured network packets using Wireshark to identify potential security issues such as unauthorized access attempts, clear-text credentials, suspicious communication, and other anomalous network activities.

Tools Required:
Wireshark — a free and open-source network protocol analyzer.

Pre-Lab Questions:

1) What is TCP, and why is it considered a connection-oriented protocol?

2) What are the functions of the bind(), listen(), and accept() methods in a TCP server program?

Post-Lab Questions:

1) What would happen if the client attempts to connect before the server starts listening, and how can this issue be resolved ?

Theory

Network packets contain information exchanged between devices over a network. Packet analysis allows a network administrator or security analyst to examine this communication and identify abnormal or potentially malicious activity.

Wireshark captures packets from a selected network interface and presents them in a human-readable format. Each packet can be examined at different protocol layers, including:

- Ethernet
- IP
- TCP / UDP
- DNS
- HTTP / HTTPS
- ARP
- ICMP
- DHCP
- TLS

Security analysis of captured packets can reveal indicators such as:

A. Unauthorized Access Attempts

Repeated connection attempts to a host or service may indicate unauthorized access or scanning.

Source IP	Destination IP	Protocol	Port
192.168.1.20	192.168.1.10	TCP	22
192.168.1.20	192.168.1.10	TCP	23
192.168.1.20	192.168.1.10	TCP	80

B. Clear-Text Credentials

Protocols such as **HTTP, FTP, and Telnet** can transmit information without adequate encryption.

For a controlled lab demonstration, traffic generated by a test service can be inspected using:

```
http
ftp
telnet
```

If sensitive information is visible directly in packet contents, it demonstrates the security weakness of transmitting credentials in clear text.

C. Suspicious Network Communication

Unexpected connections can be investigated by examining:

- Source IP
- Destination IP
- Source/Destination ports
- Protocol
- Packet frequency
- TCP flags

Experimental Setup :

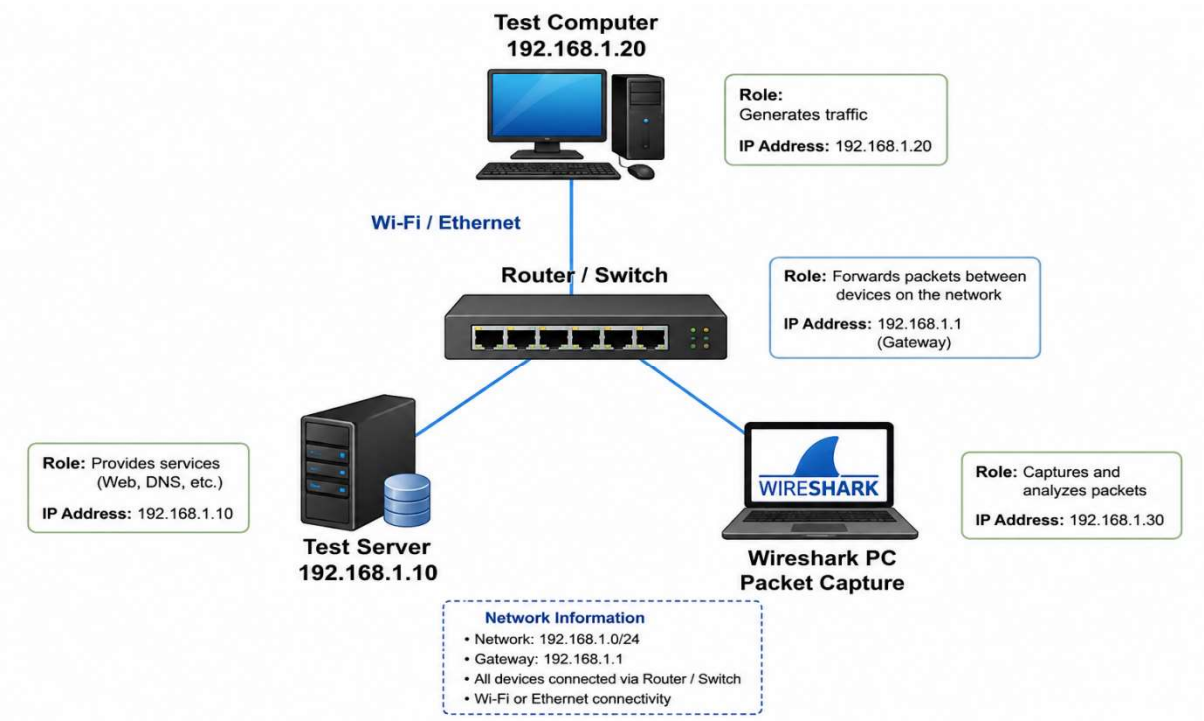


Fig 1

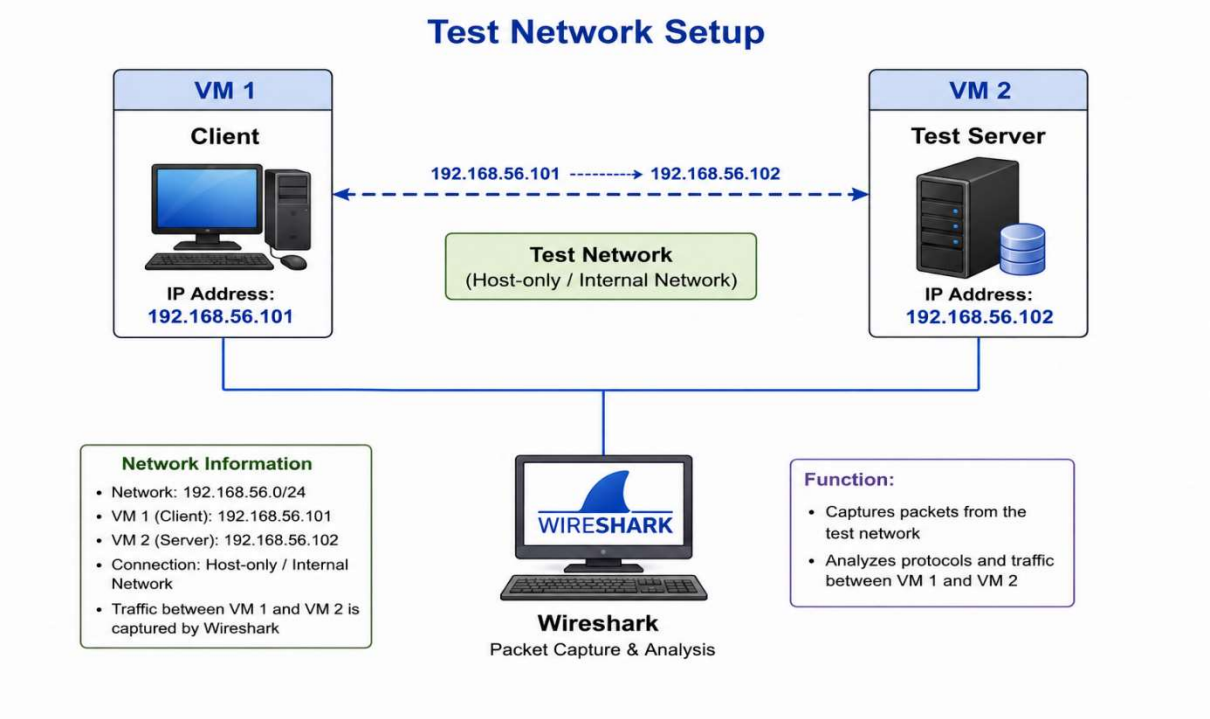


Fig 2 :

Procedure

Step 1: Install Wireshark

Install Wireshark on your laboratory computer.

During installation, allow the packet-capture component such as **Npcap** when prompted on Windows.

Step 2: Start Wireshark

Open Wireshark.

You will see available network interfaces such as:

Wi-Fi

Ethernet

Loopback

Step 3: Start Packet Capture

Double-click the appropriate interface.

For example:

Wi-Fi

Wireshark will begin capturing packets.

You should see packets similar to:

No.	Time	Source	Destination	Protocol
1	0.000	192.168.1.20	192.168.1.1	DNS
2	0.020	192.168.1.20	192.168.1.10	TCP
3	0.040	192.168.1.10	192.168.1.20	TCP

5. Generate Controlled Test Traffic

For a laboratory experiment, generate **your own traffic**.

You can:

Test A — Web Traffic

Visit a test HTTP service in your isolated lab environment.

Then use:

http

as the Wireshark display filter.

Test B — DNS Traffic

Open a website or perform a DNS lookup in your test environment.

Use:

dns

You can examine:

Query Name

Source IP

Destination IP

Response

Test C — TCP Traffic

Generate normal TCP connections between your test machines.

Use:

tcp

Observe:

SYN

SYN, ACK

ACK

Test D — ICMP

From your test client, ping your test server:

ping 192.168.56.102

In Wireshark use:

icmp

You should observe:

Echo Request
Echo Reply

Investigating Clear-Text Data

In your isolated test environment, generate dummy HTTP/FTP/Telnet traffic.

Then select the relevant packet.

You can use:

Right-click → Follow → TCP Stream

Wireshark may reconstruct the application conversation.

For a controlled demonstration, you may see data such as:

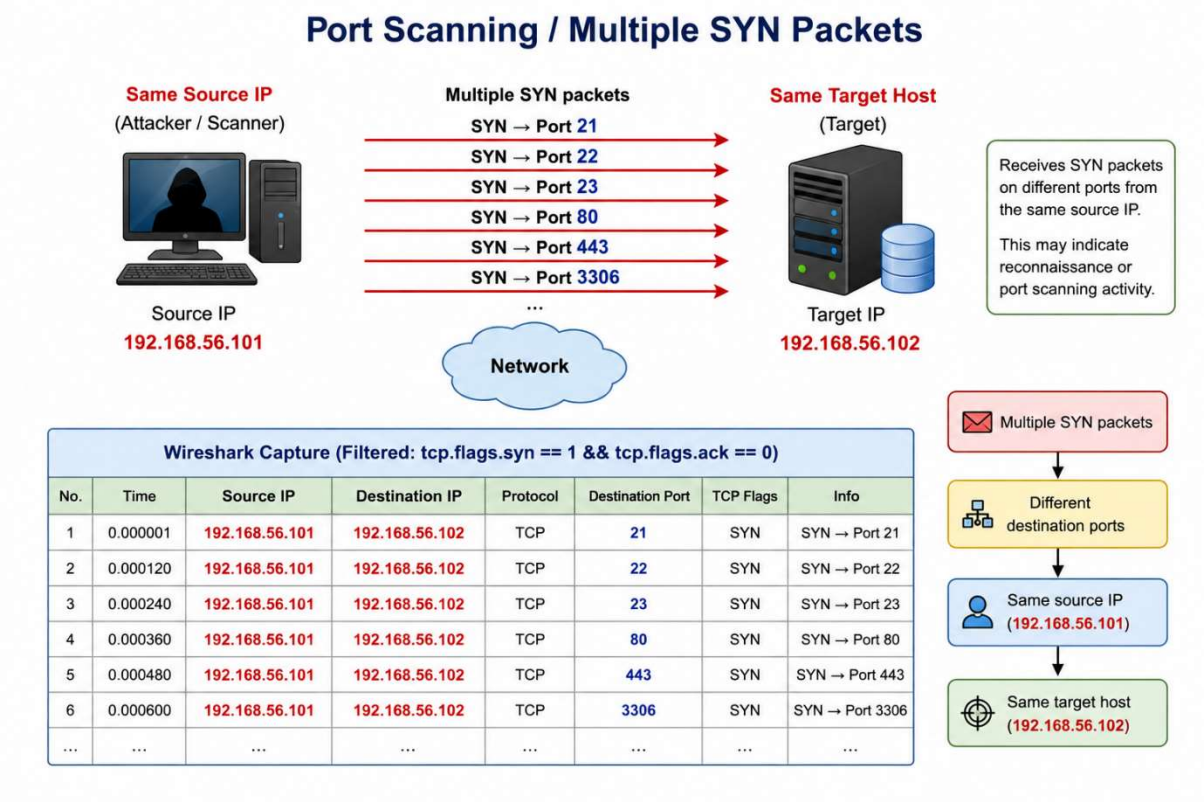
username=testuser
password=LabPassword123

This demonstrates the major security problem:

> Sensitive information transmitted over an unencrypted protocol can potentially be observed by someone capable of capturing the traffic.

Use dummy credentials only.

Investigating Possible Unauthorized Access



This could be **consistent with reconnaissance or port scanning**, but packet evidence alone does not prove malicious intent.

Therefore, record:

- Source IP
- Destination IP
- Destination port
- Protocol
- Timestamp
- Number of attempts
- TCP flags

Expected Output

Your output can be documented as follows:

Output 1 — Packet Capture

No.	Time	Source	Destination	Protocol
1	0.001	192.168.56.101	192.168.56.102	ARP
2	0.010	192.168.56.101	192.168.56.102	TCP
3	0.012	192.168.56.102	192.168.56.101	TCP
4	0.015	192.168.56.101	192.168.56.102	HTTP

Output 2 — HTTP Analysis

```
Protocol: HTTP
Source: 192.168.56.101
Destination: 192.168.56.102
Observation: Application data transmitted without encryption
Security Issue: Clear-text communication
```

Output 3 — TCP Analysis

```
Source: 192.168.56.101
Destination: 192.168.56.102

Observed:
SYN
SYN-ACK
ACK

Conclusion:
Normal TCP three-way handshake observed.
```

Output:

Conclusion:

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CO's Covered:

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ASSIGNMENT 1

Rubric	***Score***
Delivery(0-3)	
Understanding (0-3)	
Diagrams (0-3)	
Total	

Submitted On: _____

Sign: _____

ASSIGNMENT 2

Rubric	***Score***
Delivery(0-3)	
Understanding (0-3)	
Diagrams (0-3)	
Total	

Submitted On: _____

Sign: _____